

Homework Assignment 5

JPEG Image Compression & Morphological Image Processing

Digital Image Processing

Deadline: 09:00 AM, 12/15/2025

HW 5(A)

a) Given the following quantized image block in the DCT domain, you are asked to encode the AC coefficients in zig-zag order using arithmetic encoding, with the end-of-block symbol to the end of the bitstream. The probability distribution of possible quantized AC coefficients is provided below. You need to save the arithmetic code (a binary string) in A.txt. (30%)

Symbol	Probability	Range
-5	0.001	[0, 0.001)
-4	0.002	[0.001, 0.003)
-3	0.004	[0.003, 0.007)
-2	0.020	[0.007, 0.027)
-1	0.040	[0.027, 0.067)
0	0.86	[0.067, 0.927)
1	0.040	[0.927, 0.967)
2	0.020	[0.967, 0.987)
3	0.004	[0.987, 0.991)
4	0.002	[0.991, 0.993)
5	0.001	[0.993, 0.994)
\$ (EOB)	0.006	[0.994, 1.0)

Probability distribution of symbols

-9	-3	0	1	0	0	0	0
3	1	2	0	0	0	0	0
0	0	-1	0	0	0	0	0
2	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Quantized image block

HW 5(A)

- b) In this part, you are asked to manually analyze the bitstream and reconstruct the hierarchical structure according to the JPEG standard. You must parse the bitstream to identify all markers and segments. Your report should clearly show the parsed hierarchy diagram, byte offsets, and segment meanings with the required report format (see next page). Programming is not required for this problem. A copy of the CCITT T.81 standard document is provided on NTU COOL for your reference. (40%)

Required Report Format

Tables

- <Segment name>/<Segment Description>/<Offset (byte)>/<Marker (Hex)>

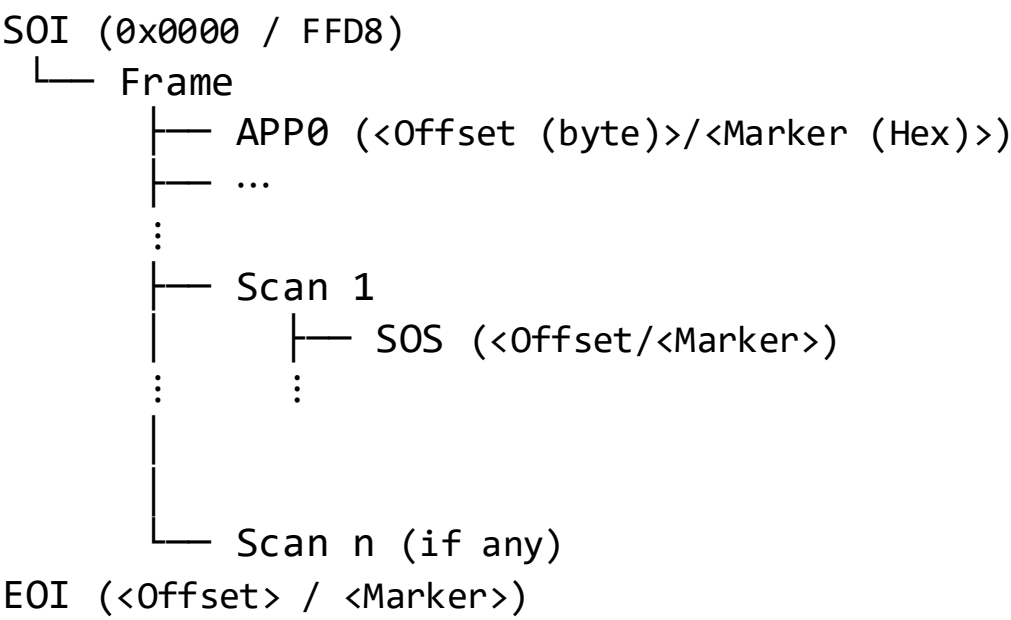
Parameter	Value
Marker	(Hex value)
Definition length	(Decimal)
... (parameters of each segment)	-

For example:

- DQT / Define quantization table / 0x0014 / FFDB

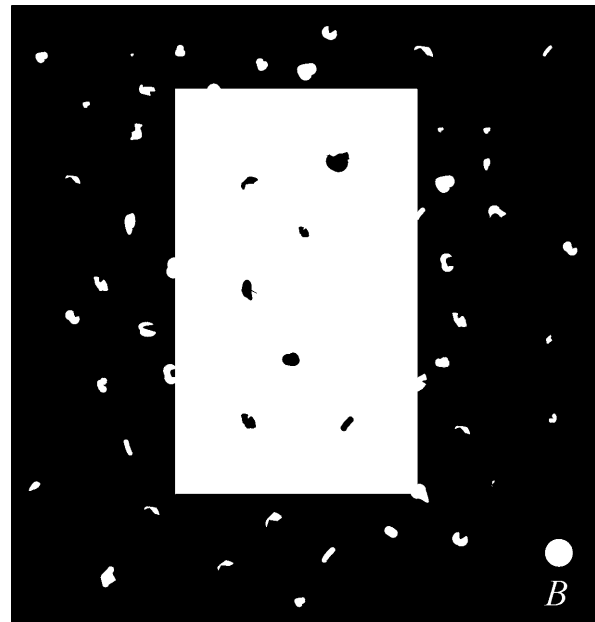
Parameter	Value
Marker	FFDB
Definition length	67
Element precision	0
Destination identifier	0
Table element (byte)	03 02 02 ... ⋮

Hierarchy Diagram



HW 5(B)

Refer to the image and the disk structuring element shown in the lower right of the image. Sketch what the sets C, D, E, and F would look like for the following sequence of operations: $C = A \ominus B$; $D = C \oplus B$; $E = D \oplus B$; and $F = E \ominus B$. Set A consists of all the foreground pixels (white), except the structuring element, B, which you may assume is just large enough to encompass any of the random elements in the image. Note that the sequence of operations above is simply the opening of A by B followed by a closing of the result by B. (30%)



noisy_rectangle.tif

Grading Policy

- You should provide a detailed description of the computation procedures.
- The score will be based on the quality (correctness) of your results and analysis.
- Your computer code will be tested by TA.
- Total points: 100.

Assignment Requirements

- In Part (a) of HW 5(A), you should **not** use built-in functions for encoding.
- In HW 5(B), you should **not** use built-in functions for morphological operations.
- Provided files
 - Image_hex.txt
 - noisy_rectangle.tif
- Set your directory structure as follows:
 - hw5_<your ID>/
 - <program files>
 - report.pdf
 - results/
 - A.txt
 - <Image of program output>

Software Package Allowed

- Python 3.8+
- Standard Python library
- Numpy
- Opencv-python

Assignment Submission Requirements

- Submit to **NTU COOL**
- Do NOT copy homework (code, report, results, etc.) from others